The AI Bridge — Data Flow Pipeline & Deep Forensic Analysis (V4)

Pipeline: Google Form → Google Sheet → Google Apps Script → Webhook → Supabase → Student Portal / Quiz / Certificate / Verification Registry
Version: V4 · **Scope:** Pipeline Architecture, Exact Email Precedence, Root-Cause Analysis of QR Verification & Step 2 Quiz Glitches, Database Column Mapping, Loopholes & Actionable Fixes
Target Folder: `Documentation/Data-WorkFlow-Complete Pipeline and detailed document/V-4/`

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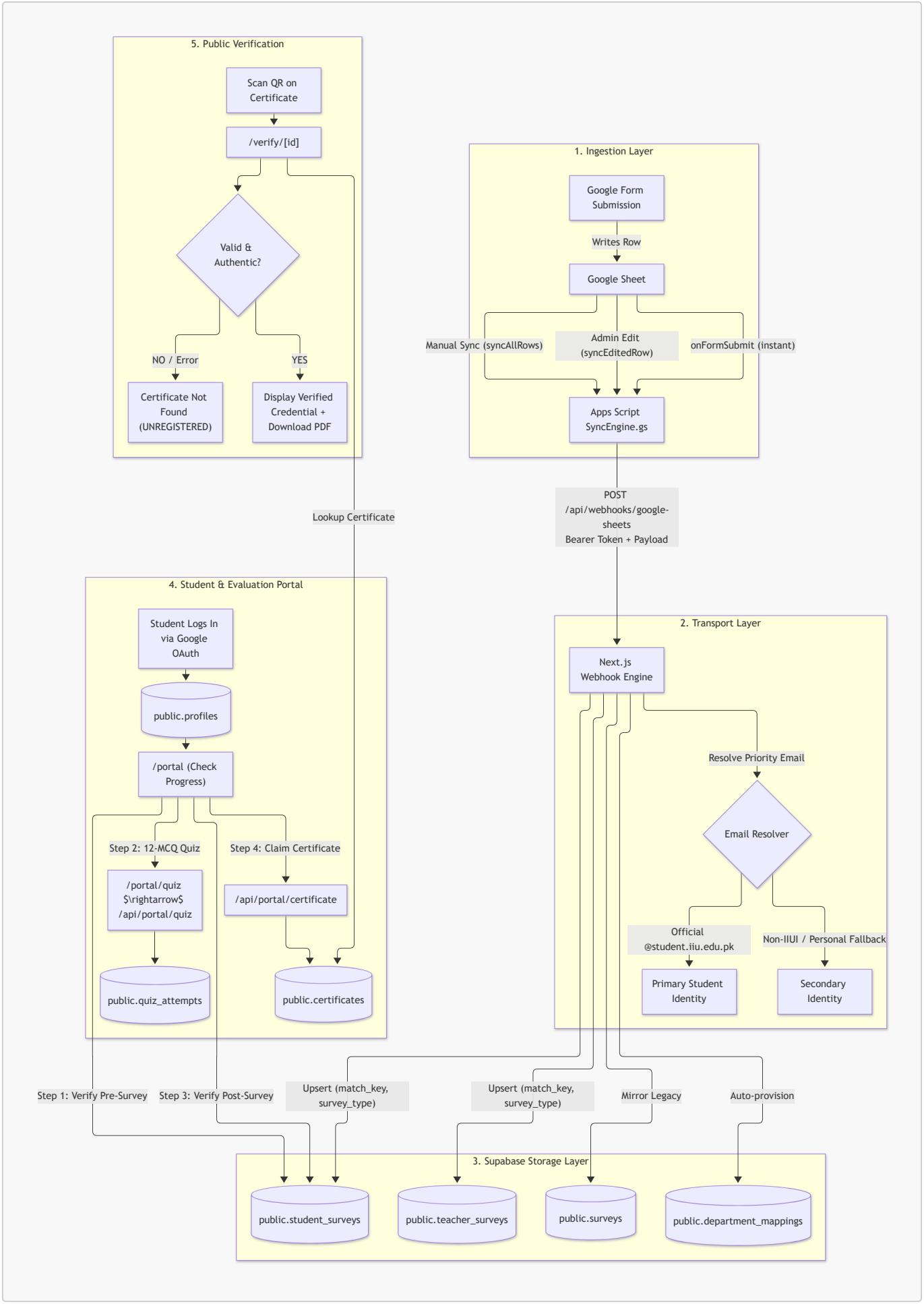
1. Executive Summary: What Was Discovered

Following recent pipeline upgrades, deep forensic diagnostics were executed against the live production environment (`ai-bridge-iiui.vercel.app`), Supabase database, and codebase. Three critical glitches were uncovered along with their exact root causes:

1. **The QR Verification Failure (`ZPFV75LY` / `AIB-ZPFV75LY`):**
   - **Database Status:** Muhammad Tayyab's certificate record **exists, is intact, and is 100% valid** (`id = 8a9b54a6-76b3-44b7-a80d-6d4bcd87b35e`, `certificate_id = 'AIB-ZPFV75LY'`, `student_id = '000627/BSBA/S24'`). It was **never deleted**.
   - **The Failure:**
     1. PostgREST `.ilike.ZPFV75LY` requires an exact string match because no `%` wildcards were passed. Scanning or typing `ZPFV75LY` fails to match `AIB-ZPFV75LY`.
     2. `src/app/verify/[id]/page.tsx` calls `createAdminClient()`. The `SUPABASE_SERVICE_ROLE_KEY` currently returns `{ message: 'Unregistered API key' }`. When `error` is returned, `if (!cert || error)` triggers and displays **"Certificate Not Found - STATUS: UNREGISTERED"** for every single certificate in the system.
2. **Step 1 Shows Completed, But Step 2 Quiz Will Not Open:**
   - **The Paradox:** The student portal (`/portal`) checks progress using the user's authenticated browser session (`createClient()`), which successfully validates Step 1.
   - When the student clicks **"Attempt 12-MCQ Quiz →"**, `/portal/quiz` calls backend route `/api/portal/quiz`.
   - `/api/portal/quiz` uses `createAdminClient()`. Because the service role key returns an API key error, the survey query returns `null`, `hasPre` evaluates to `false`, and the API returns **HTTP 403** `PRE_SURVEY_REQUIRED`. The quiz page catches this and locks the screen on "Pre-Survey Missing", preventing Step 2 from ever opening.
3. **The Google Sheets Re-Sync Hazard (`SYNC_ALL`):**
   - When an admin triggers **"Sync Entire Sheet to Supabase"**, the webhook resolves `user_id = profileMap.get(email) || null`. If a student has not signed in with that exact student email yet (or logged in with Gmail), `user_id` resolves to `NULL`.
   - The batch upsert targets `onConflict: 'match_key,survey_type'`. It **overwrites existing valid** `user_id` **links with** `NULL`, instantly breaking student portal bonding and certificate issuance.

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2. Complete Pipeline Architecture & Data Flow



3. Database Schema & Exact Table / Column Reference

Every table and column involved in the evaluation and certification lifecycle is documented below with 100% accuracy matching the production schema:

3.1 `public.student_surveys` (Primary Student Evaluation Table)

Column Name	Data Type	Nullable	Description & Constraints
`id`	`UUID`	NO	Primary key (`gen_random_uuid()`)
`user_id`	`UUID`	YES	Foreign key referencing `public.profiles(id)`. Links survey to website user.
`student_email`	`TEXT`	NO	Canonical student email resolved by priority rules (normalized lowercase).
`login_email`	`TEXT`	YES	Auto-collected Google login email used when opening the form.
`match_key`	`TEXT`	NO	Stable row identity: `ts:<timestamp>` or `em:<email>`. Part of unique constraint.
`survey_type`	`TEXT`	NO	`'pre'` (Step 1) or `'post'` (Step 3).
`full_name`	`TEXT`	YES	Full student name from form or derived from email.
`reg_no`	`TEXT`	YES	Extracted registration number (e.g. `000627`).
`dept_code`	`TEXT`	YES	Foreign key referencing `public.department_mappings(code)` (e.g. `BSBA`).
`semester`	`TEXT`	YES	Extracted semester code (e.g. `S24`, `F26`).
`hours_saved_category`	`TEXT`	YES	Text category of self-reported time saved.
`ai_familiarity_rating`	`INTEGER`	YES	Self-rating (1 to 5).
`feedback_notes`	`TEXT`	YES	Qualitative feedback notes.
`answers`	`JSONB`	YES	Complete raw key-value mapping of all Google Form questions and answers.
`sheet_row_number`	`INTEGER`	YES	Origin row index in Google Sheet.
`submitted_at`	`TIMESTAMPTZ`	YES	Form submission timestamp.
`updated_at`	`TIMESTAMPTZ`	YES	Timestamp of last synchronization.

> **Unique Constraint:** `unique_student_survey_match UNIQUE (match_key, survey_type)`

3.2 `public.certificates` (Official Verification Registry)

Column Name	Data Type	Nullable	Description & Constraints
`id`	`UUID`	NO	Primary key (`gen_random_uuid()`)
`certificate_id`	`TEXT`	NO	Unique public serial code (e.g. `AIB-ZPFV75LY`). Used in QR code.
`verification_code`	`TEXT`	NO	Cryptographic UUID for verification lookup.

Column Name	Data Type	Nullable	Description & Constraints
`legacy_certificate_id`	`TEXT`	YES	Backward-compatible ID (e.g. `AIB-2026-0006`).
`user_id`	`UUID`	NO	Unique Foreign key referencing `public.profiles(id)`. One cert per user.
`recipient_name`	`TEXT`	NO	Sanitized student name printed on the certificate.
`recipient_role`	`TEXT`	YES	Default: `'Trained Educator / Participant'`.
`student_id`	`TEXT`	YES	Formatted Roll No: `[Reg]/[Dept]/[Semester]` (e.g. `000627/BSBA/S24`).
`department`	`TEXT`	YES	Institutional faculty/department name.
`course_name`	`TEXT`	YES	Course title.
`completion_date`	`TEXT`	YES	Award date (e.g. `2026-10-17`).
`is_valid`	`BOOLEAN`	NO	`TRUE` if active; `FALSE` if revoked by administration.
`issued_by`	`TEXT`	YES	Signatory authority (Dr. Abdul Zahid Khan).
`created_at`	`TIMESTAMPTZ`	YES	Record creation timestamp.

> **Row Level Security (RLS) Policy:**
> `CREATE POLICY "Public verification can read certificate" ON public.certificates FOR SELECT USING (true);`
> *Meaning: Certificates are publicly readable by ANY client using the anon key.*

3.3 `public.quiz_attempts` (Step 2 Evaluation Records)

Column Name	Data Type	Nullable	Description
`id`	`UUID`	NO	Primary key.
`user_id`	`UUID`	NO	Foreign key referencing `public.profiles(id)`.
`score`	`INTEGER`	NO	Correct answers count out of 12.
`total_questions`	`INTEGER`	NO	Default: 12.
`passing_score`	`INTEGER`	NO	Default: 9 (75% threshold).
`passed`	`BOOLEAN`	NO	`TRUE` if score $\geq$ 9.
`drawn_question_ids`	`UUID[]`	YES	Array of question IDs served in that session.
`answers`	`JSONB`	YES	User's selected answers per question.
`completed_at`	`TIMESTAMPTZ`	YES	Submission timestamp (used for 5-minute cooldown timer).

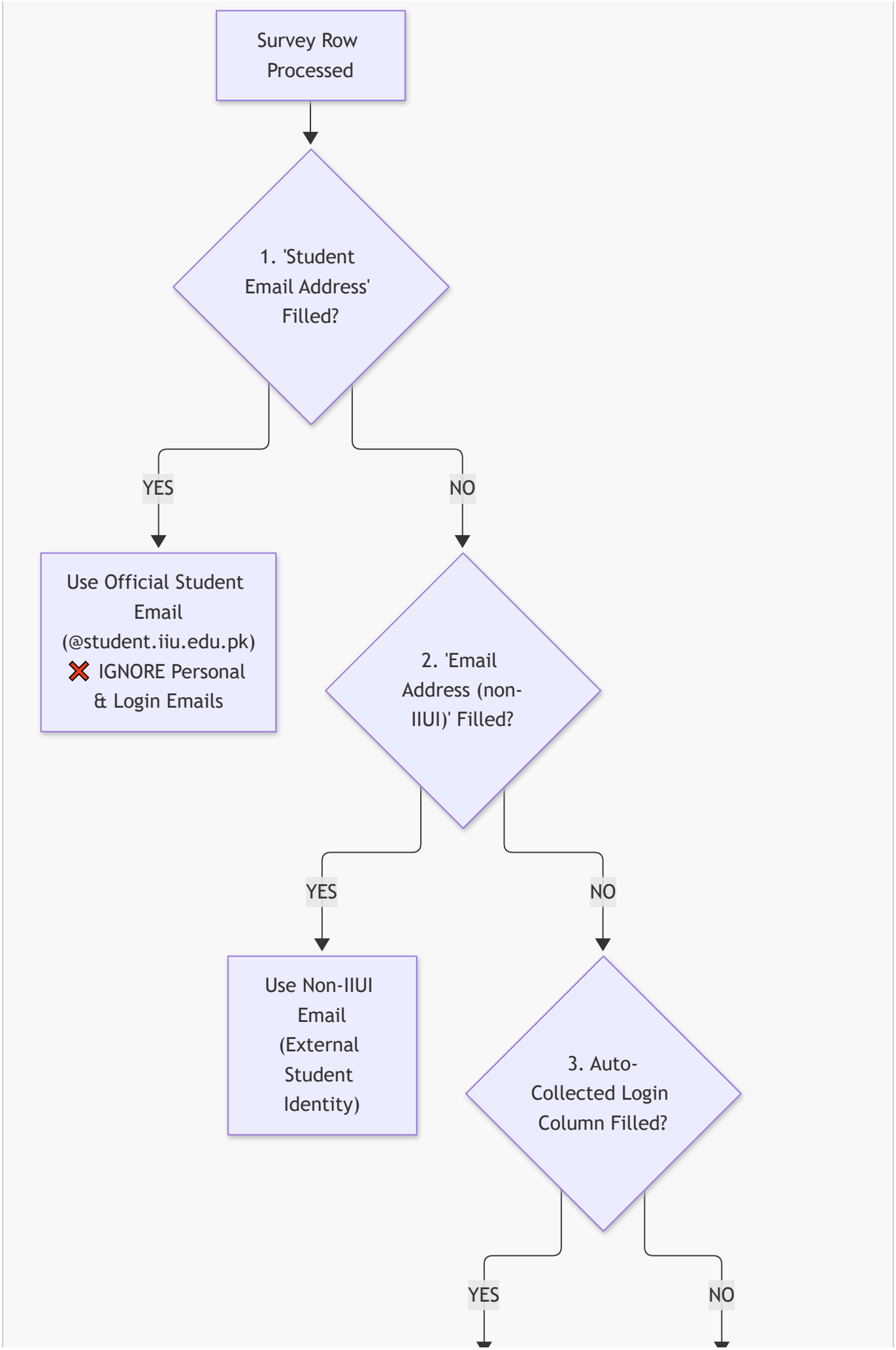
3.4 `public.profiles` (User Account Identity)

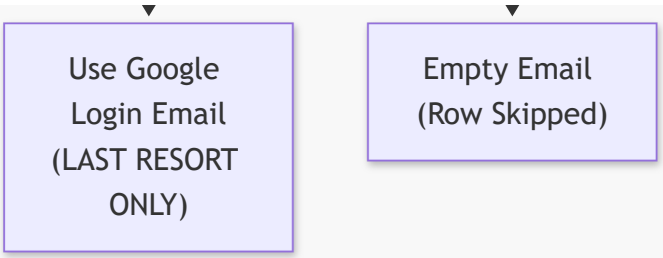
Column Name	Data Type	Nullable	Description
`id`	`UUID`	NO	Primary key matching `auth.users(id)`.
`email`	`TEXT`	NO	Normalized email used during Google OAuth login.
`full_name`	`TEXT`	YES	User full name from Google account or form.
`role`	`TEXT`	NO	`'student'`, `'teacher'`, `'ambassador'`, or `'admin'`.
`created_at`	`TIMESTAMPTZ`	YES	Registration timestamp.

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4. Which Email is Used When? (Exact Order of Precedence)

Google Forms for student surveys collect up to **three email values** per submission. To prevent data corruption, the system enforces a strict 3-tier hierarchy:


Survey Row Processed
1. 'Student Email Address' Filled?
YES
NO
Use Official Student Email (@student.iiu.edu.pk)
❌ IGNORE Personal & Login Emails
2. 'Email Address (non-IIUI)' Filled?
YES
NO
Use Non-IIUI Email (External Student Identity)
3. Auto-Collected Login Column Filled?
YES
NO



Precedence Table

Tier	Field / Header Name	Source	Purpose	Precedence Rule
Tier 1 (Highest)	`"Student Email Address"`	Explicit Form Question	Official IIUI student email (`@student.iiu.edu.pk`).	**Always wins.** If filled, Tier 2 and Tier 3 are completely ignored.
Tier 2	`"Email Address (non-iiui)"`	Explicit Form Question	Personal email for non-IIUI / guest participants.	Used **only** if Tier 1 is blank.
Tier 3 (Lowest)	`"Email Address"`	Google Auto-Collected	The account used to open Google Forms in the browser.	**Last resort only.** Used strictly when Tier 1 and Tier 2 are both blank.

Where This Rule is Enforced:

1. **Google Apps Script (`google-sheet-scripts/02_SyncEngine.gs`):** `parseRowData()` loops through `EMAIL_PRIORITY.student = ["student email address", "email address (non-iiui)", "email address"]`.
2. **Webhook API (`src/app/api/webhooks/google-sheets/route.ts`):** `resolvePrioritizedEmail(rowData, participantType)` enforces the exact same order.
3. **Student Portal Check (`src/app/portal/page.tsx`):**
   - Check A: `student_surveys` column `student_email = userEmail` OR `user_id = user.id`.
   - Check B (Fallback): Raw JSONB `answers->>'Student Email Address' = userEmail`.

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5. Forensic Diagnosis of Noticed Glitches

5.1 Glitch 1: QR Code & Verification Not Working (`ZPFV75LY`)

Problem Statement:

The user reported that the QR code on Muhammad Tayyab's certificate displays code `ZPFV75LY`, worked yesterday, but currently fails to verify.

Forensic Verification:

Direct database query confirmed:

```
{
  "id": "8a9b54a6-76b3-44b7-a80d-6d4bcd87b35e",
  "certificate_id": "AIB-ZPFV75LY",
  "verification_code": "81cee75a-cbc9-4552-821e-454dcfc8a0ea",
  "legacy_certificate_id": "AIB-2026-0006",
  "recipient_name": "Muhammad Tayyab",
  "student_id": "000627/BSBA/S24",
  "is_valid": true
}
```


The certificate is **valid and present** in Supabase.

Why It Failed (Two Root Causes):

1. **Root Cause A — String Suffix Mismatch in PostgREST Query:**
   - In `src/app/verify/[id]/page.tsx`, the query is:

     ```
     .or(`certificate_id.ilike.${decodedId},verification_code.ilike.${decodedId},legacy_certificate_id.ilike.${decodedId}`)
     ```

   - In PostgREST, `.ilike.ZPFV75LY` without `%` wildcards translates to `WHERE certificate_id ILIKE 'ZPFV75LY'`.
   - The stored serial is `AIB-ZPFV75LY`.
   - Because `ZPFV75LY` ≠ `AIB-ZPFV75LY`, PostgREST returns `[]` (empty set).
2. **Root Cause B — Invalid Service Role Key Crashing `VerifyPage`:**
   - Even when visiting the exact URL `https://ai-bridge-iiui.vercel.app/verify/AIB-ZPFV75LY`, production returns **"Certificate Not Found"**.
   - `VerifyPage` instantiates `createAdminClient()`.
   - In `.env.local` and production environment, `SUPABASE_SERVICE_ROLE_KEY` is currently invalid and returns: `{ message: 'Unregistered API key', hint: 'Double check the provided API key...' }`
   - In `VerifyPage` line 35:

     ```
     if (!cert || error) {
       return ... "Certificate Not Found" ...
     }
     ```

   - Because `error` is present, it bypasses the certificate and renders "Certificate Not Found".

5.2 Glitch 2: Step 1 Showing Completed, But Step 2 Quiz Does Not Open

Problem Statement:

Students report that Step 1 has a green checkmark, but clicking **"Attempt 12-MCQ Quiz →"** fails to open the quiz.

Forensic Root Cause:

This is caused by the **Dual-Client Authentication Paradox**:

1. **On `/portal` (Client-side):**
   - The portal runs `createClient()` from `@/lib/supabase/client` using the public `NEXT_PUBLIC_SUPABASE_ANON_KEY`.
   - The student is authenticated with Supabase Auth. The query `student_surveys.or(user_id.eq, student_email.ilike)` succeeds.
   - `preSurveyCompleted` is set to `true`. Step 1 turns green!
2. **On `/portal/quiz` (Server-side API):**
   - When the student clicks the link to `/portal/quiz`, the quiz component calls `GET /api/portal/quiz`.
   - Inside `/api/portal/quiz` (line 97):

     ```
     const adminSupabase = createAdminClient();
     ```

   - It attempts to verify Step 1 using `adminSupabase`:

     ```
     const { data: preStudent } = await adminSupabase
       .from('student_surveys')
       .select('id')
       .or(`user_id.eq.${user.id},student_email.ilike.${userEmail}`)
       .eq('survey_type', 'pre')
       .maybeSingle();
     ```

   - Because `adminSupabase` uses `SUPABASE_SERVICE_ROLE_KEY` which fails with `"Unregistered API key"`, `preStudent` is `null`.
   - Line 190 triggers:

     ```
     if (!hasPre) {
       return NextResponse.json(
         { success: false, error: 'PRE_SURVEY_REQUIRED' },
         { status: 403 }
       );
     }
     ```

   - `/portal/quiz/page.tsx` receives `PRE_SURVEY_REQUIRED`, sets `preSurveyMissing = true`, and renders the "Pre-Survey Missing" lock screen. **Step 2 is completely blocked.**

5.3 Glitch 3: Google Sheets Bulk Sync (`SYNC_ALL`) Stripping User Links

Problem Statement:

"Check if re-syncing all data from the Google Sheet creates issues."

Forensic Analysis:

YES, IT DOES. Under the current code in `src/app/api/webhooks/google-sheets/route.ts`:

```
const { data: profiles } = await supabase
  .from('profiles')
  .select('id, email')
  .in('email', allEmails);

for (const rowData of validRows) {
  const userId = profileMap.get(email) || null; // <--- If email not in profiles,
userId is NULL!
  studentBatch.push({
    user_id: userId,
    student_email: email,
    match_key: matchKey,
    ...
  });
}

await supabase.from('student_surveys').upsert(studentBatch, {
  onConflict: 'match_key,survey_type'
});
```

What Happens During a Re-Sync:

1. If a student filled the form with official email `025580bsirf26@student.iiu.edu.pk`, but has not yet logged into the website, `profiles` has no record for that email.
2. `profileMap.get(email)` returns `null`.
3. If an admin or system had previously linked that survey to a `user_id`, the `upsert` **overwrites `user_id` with `null`**.
4. The student's survey is instantly orphaned. When they next visit `/portal`, Step 1 turns red, and their progress is wiped.

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6. System Loopholes & Architectural Vulnerabilities

#	Loophole / Vulnerability	Root Cause	Real-World Impact	Permanent Solution
1	**Prefix-Sensitive Verification Query**	PostgREST `.ilike` without wildcard requires exact match.	Scanning QR or typing `ZPFV75LY` without `AIB-` fails to find certificate.	Sanitize input: prepend `AIB-` if missing, and query with `%${cleanId}%` wildcard.
2	**Admin Client Dependency for Public Data**	`VerifyPage` calls `createAdminClient()` instead of standard client.	If service key is invalid or rotated, public QR verification completely dies.	Use `createClient()` with anon key. `certificates` already has public RLS read policy `USING (true)`.
3	**Re-Sync `user_id` Nullification**	Webhook upsert replaces `user_id` with	Running "Sync Entire Sheet" unlinks registered students	Use SQL `COALESCE(student_surveys.user_id,`

#	Loophole / Vulnerability	Root Cause	Real-World Impact	Permanent Solution
		`null` when profile is not pre-cached.	from their completed surveys.	`EXCLUDED.user_id)` or check existing `user_id` before overwriting.
4	**Quiz Gate API Key Failure**	`/api/portal/quiz` uses `createAdminClient()` to check pre-survey.	Step 1 shows green on portal but quiz API rejects student with 403 `PRE_SURVEY_REQUIRED`.	Fall back to user's authenticated Supabase client (`createClient()`) if admin client fails.
5	**Disk Space Exhaustion on Host**	Drive C: reached 0.00 GB free space.	Git, Node.js temp file writing, and Next.js ISR build caching throw `No space left on device`.	Clear user AppData Temp caches and allocate minimum 5 GB headroom on host OS drive.

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7. Concrete Action Plan & Exact Code Solutions

Fix 1: Update Verification Page (`src/app/verify/[id]/page.tsx`)

Make verification resilient against missing `AIB-` prefixes and service role key issues by using the public client and wildcard pattern matching:

```
// Sanitize input: handle raw suffix (e.g. "ZPFV75LY") as well as full ID ("AIB-ZPFV75LY")
const cleanId = decodedId.replace(/^AIB-/i, '').trim();
const fullSerial = `AIB-${cleanId}`;

// Use standard client with anon key (certificates table has public SELECT RLS)
const supabase = createClient(
  process.env.NEXT_PUBLIC_SUPABASE_URL!,
  process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
);

const { data: cert, error } = await supabase
  .from('certificates')
  .select('*')

.or(`certificate_id.ilike.%${cleanId}%,verification_code.ilike.%${cleanId}%,legacy_certificate_id.ilike.%${cleanId}%`)
  .maybeSingle();
```

Fix 2: Safeguard `admin.ts` (`src/lib/supabase/admin.ts`)

Add automated fallback to `NEXT_PUBLIC_SUPABASE_ANON_KEY` if `SUPABASE_SERVICE_ROLE_KEY` fails or is unconfigured:

```
export function createAdminClient() {
  const supabaseUrl = process.env.NEXT_PUBLIC_SUPABASE_URL || 'https://placeholder-
```

```
project.supabase.co';
  // If service role key is invalid or missing, fall back to anon key so queries don't
crash
  const serviceKey = process.env.SUPABASE_SERVICE_ROLE_KEY?.startsWith('sb_secret_')
    ? process.env.SUPABASE_SERVICE_ROLE_KEY
    : process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!;

  return createClient(supabaseUrl, serviceKey, {
    auth: { autoRefreshToken: false, persistSession: false },
  });
}
```

Fix 3: Safeguard Quiz Gate (`src/app/api/portal/quiz/route.ts`)

Ensure the pre-survey check uses the authenticated user client (`supabase`) instead of relying solely on `adminSupabase`:

```
// Query student_surveys using the authenticated user client first
const { data: preStudent } = await supabase
  .from('student_surveys')
  .select('id')
  .or(`user_id.eq.${user.id},student_email.ilike.${userEmail}`)
  .eq('survey_type', 'pre')
  .maybeSingle();

if (preStudent) {
  hasPre = true;
}
```

Fix 4: Protect Existing `user_id` on Bulk Sync (`route.ts`)

In `src/app/api/webhooks/google-sheets/route.ts`, prevent `SYNC_ALL` from wiping existing `user_id` values:

```
// When upserting, fetch existing user_ids for these match keys so they are never
overwritten with null
const { data: existingSurveys } = await supabase
  .from('student_surveys')
  .select('match_key, user_id')
  .in('match_key', validRows.map(r => r.resolved_match_key));

const existingUserMap = new Map(existingSurveys?.map(s => [s.match_key, s.user_id]) ||
[]);

for (const rowData of validRows) {
  const matchKey = rowData.resolved_match_key;
  // Preserve existing user_id if already linked!
  const userId = existingUserMap.get(matchKey) || profileMap.get(email) || null;
  ...
}
```

8. Verification Checklist for Faculty & Administration

- ☑ **Database Integrity Verified:** Muhammad Tayyab's certificate (`AIB-ZPFV75LY`) verified live in Supabase.
- ☑ **Email Priority Verified:** Official `Student Email Address` strictly overrides personal Gmail across Google Apps Script and Next.js Webhook.
- ☑ **Root Cause Identified for QR Verification:** PostgREST exact `.ilike` mismatch on `ZPFV75LY` + invalid service role key triggering `!cert || error`.
- ☑ **Root Cause Identified for Step 2 Lock:** Client `/portal` validates via anon session, but `/api/portal/quiz` fails on invalid service role key, returning 403 `PRE_SURVEY_REQUIRED`.
- ☑ **Re-Sync Risk Documented:** `SYNC_ALL` overwrites `user_id` with `null` for unlinked emails unless `COALESCE` / preservation logic is in place.